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Conceptual Design in Informal Metalworking Microenterprises of Tanzania

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Abstract: Product design is a key aspect of human intelligence and creativity, attracting not only experts but also people without any formal design training. Although numerous people in developing countries design and manufacture products in metalworking microenterprises in the informal sector, there is still little knowledge about their design process. This paper aims to fill this gap in design knowledge. We aim to investigate the design processes in metalworking microenterprises in the informal sector of Tanzania. In particular, we aim to explore how these microenterprises identify consumer needs and requirements, how they determine the specifications for the product, how they generate and evaluate alternative product concepts, and how they define product details. To address these aims, semistructured interviews were carried out in metalworking microenterprises operating in the informal sector of Tanzania. The findings reveal many facets of their design processes, providing a sound basis upon which design methods and tools can be developed to support their design activities.

Keywords: conceptual design; design practice; informal sector; microenterprises; sustainable development; developing countries

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1. Introduction

People around the world are increasingly living in artificial designed environments proliferated with a wide range of designed products, such as electronic gadgets and vehicles, as well as relatively simple products such as kitchen utensils, bicycles, furniture, etc. Design activities and their outcomes profoundly affect our lives and well-being, while impacting social and environmental sustainability, plus the economic growth of organisations involved in design, manufacturing, and other life cycle phases of products [1]. Humans have designed many kinds of products in the past; they are doing this presently and will engage in design activities in foreseeable future as well. Whilst people with formal design education might have more abilities to design products, those lacking formal education also have abilities to design a broad range of products [2–4].

Design processes offer opportunities to design products that satisfy intended requirements while minimising unintended consequences [5]. As such, the scientific field of "design research" not only aims to gain an in-depth understanding of design processes, but also aims at their improvement [1]. As design is a context-sensitive activity [2,6], there is a critical need to investigate design processes in a broad range of contexts representing various social, economic, and cultural characteristics.

However, extant design research has predominantly been undertaken in some specific contexts, typically those found in developed countries and relatively wealthy regions of the world. This has generated limited design knowledge in a narrow range of contexts, while restricting our understanding of design phenomena in the resource-constrained

settings observed in developing countries [2]. This biased contextual focus of extant design research is not just unhealthy for the growth of design research, but also hinders practitioner ability to learn from design phenomena in various contexts and to design products and services for addressing various global challenges. It is therefore crucial to investigate design activities in developing countries, including design activities in informal metalworking microenterprises [2].

Because there are many socioeconomic and cultural differences between organisations in Western countries and those of the developing world, there are large disparities in their approaches to design and how they design products [2]. Knowledge about design activities in informal metalworking firms in the developing world is needed for developing tailored training programmes and methods to support them in designing more successful and sustainable products.

Enterprises in the informal sector provide income-generation opportunities to nearly one-third of the nonagricultural global workforce [7]. Previous research in this field suggests that microenterprises make up a major part of the informal sector. In general, an organization is considered to be a microenterprise if it has no more than 10 employees, and if the majority of its employee are engaged in the manufacturing of products [8,9].

Design research, presently, is conducted largely in developed countries [10], in organizations that are highly dissimilar to informal metalworking firms in the developing world [8,11]. Thus, there is lack of systematic knowledge about design activities in such metalworking microenterprises. This research aims to address this gap in design knowledge. We address this overall aim by attempting to answer the following two broad research questions, namely (a) how do informal metalworking firms in developing countries elicit client needs and identify requirements for products? (b) How do these microenterprises develop design concepts and define detail designs, including materials, dimensions, etc.? To answer these two research questions, we conducted interviews in 24 microenterprises in the informal sector in some selected regions of Tanzania.

Following the above introduction, the remaining paper is organised as follows. Section 2 reviews the related literature, highlights the need to investigate design activities in a broad range of socioeconomic contexts, discusses the role of the informal sector in developing countries, and identifies the crucial need to explore design activities in informal metalworking microenterprises in developing counties. Section 3 provides the details of the research method, presenting information on sampling, data collection, and data analysis. Sections 4 and 5 present the findings of a qualitative analysis of interview data collected from 24 informal microenterprises. Finally, Section 6 discusses the findings, along with concluding remarks and limitations of this study.

2. Background Literature

This section presents the literature review on the design process and design research, specifically in the context of product design in informal sectors. This review was conducted using a semisystematic approach [12]. An author of this work has previously published an integrative literature review in the domain of design and poverty, specifically on role of poor people in formal and informal design sectors [2]. This integrative literature review guided the semisystematic literature review in this work. This ensured that appropriate literature is accurately covered to be able to answer the research questions.

2.1. Design Process and Design Research

Designed products are recognized as one of many important means for a society to advance its economic and social well-being [1,5]. As individuals, we appreciate a well-designed product, and respond to it by selecting it from available products [1]. The ability of individual entrepreneurs and the industry at large to design efficient, inspirational, and aesthetic products is therefore crucial. The ability to design products is recognised as one of several forms of creativity and a fundamental aspect of human intelligence [3,4]. Consequently, it is important to understand what people do when they exercise design ability [13].

In design research, in order to fully understand how people actually design products, empirical data on the design process are collected by using a variety of methods, including interviews [3,10]. Some characteristics of design processes have been widely observed. It is commonly accepted that the design process is iterative [14,15] and that identification of consumer needs, the generation and evaluation of concepts, and defining of product details are important activities in a design process [2,14,16].

A large body of research has demonstrated that the design process provides a maximum scope to improve, modify, and create new products because many crucial decisions are made in this very process [3,5,13]. Deeper scientific knowledge on the design processes is needed to develop design knowledge [1,13]. Several authors argue that generating design knowledge requires investigating design processes in a variety of settings, covering a broad range of sociocultural and economic environments in which the products are designed [17,18]. However, despite the prevalence of design activity, extant design research is predominantly focused on individuals and enterprises in developed countries, thus investigating design processes in some specific socioeconomic and cultural settings [8,10]. This narrow focus of design research has constrained the generation of design knowledge by limiting its focus [8,10]. Addressing these gaps in the existing design knowledge necessitates that design research be undertaken in a range of fields, including, among others, informal sector microenterprises in developing countries. Such microenterprises are distinctly different on many dimensions from enterprises in developed countries.

2.2. Informal Sector

Low-income people in developing countries may produce a variety of products, such as furniture, utensils, and common domestic goods. These activities of making products create jobs, supporting them to satisfy their unmet or under-served needs [2].

In developing countries, people generally work either in the informal or formal sectors. A person working in the formal sector has a formal contract with the firm owner or employer, typically receives a decent monthly salary, and generally has access to a social security system [2]. In contrast, a person working in the informal sector typically lacks legal contract with the firm owner or employer, lacks a well-ordered work environment, has an irregular work duration, typically receives a low and irregular income, and has no access to social security [2,19].

As mentioned earlier, researchers from sociology, anthropology, and economics, as well as from other disciplines, have extensively studied the informal sector [18,20]. Furthermore, in recent years, studies have been carried out on firms operating in the informal sector [19,21,22]. However, research devoted to this subject is relatively minor as there is widespread commonness of informal firms in developing countries. Recent studies suggest that the contribution of the informal sector to the GDP in developing countries is about 40–60% [23]. The informal sector employs just under one-third (31.5%) of the nonagricultural workforce around the world [2]. As such, the informal sector represents a large segment of the total global economy. For some authors, informality is a "voluntary" choice [24], whereas, for others, informality is an "exclusion" [25].

Some scholars have examined the informal sector as a labour phenomenon [22,26], examining its role in creating jobs for numerous low-income people in developing countries, as well as its failure in providing access to various benefits, including social security. Other scholars have dealt with the challenge of how to describe and define it, its relationship with other parts of the economy, and how it affects growth [7,27]. In addition, some researchers have studied the informal sector as microenterprises, which can benefit from the provision of technology and funding [28]. These studies suggest that microenterprises make up a significant fraction of the informal economy, and most of these firms manufacture products [8,9]. To sum up, the informal sector is a significant part of the total economy in developing countries. However, design researchers have given little or no attention to this sector, despite the presence of product design activities in this sector [2].

2.3. Product Design in Informal Microenterprises

Researchers generally agree on the main traits of manufacturing microenterprises operating in the informal sector. They typically use readily available tools and equipment, are labour-intensive, depend on family ownership, rely on competitive and unregulated markets, and are small-scale [2,29]. They face many constraints, such as lack of access to financial and material resources, weak infrastructure, weak marketing and organisational resources, and absence of specialised knowledge and skills [8,29]. Despite these constraints, studies have suggested that informal microenterprises have the ability to design products [8,9,30–34].

Despite the acceptance of design ability in microenterprises in the informal sector, there is little or no research on their design activities. This can partly be clarified by the perspectives of the studies undertaken in the informal sector. These studies are generally undertaken from an economic perspective and consider design activities of the microenterprises as a "black box" [2,8,9,34]. A recent study has identified the need to investigate design activities in informal microenterprises, including metalworking informal microenterprises [2].

3. Method

In general, design research studies use the following research methods: (a) design experiments; (b) questionnaire analyses (online and offline); (c) interviews (online, physical, or telephone); (d) diary studies; and (e) observational studies in real settings.

In this work, we have used interviews as a research method to collect data. These interviews were conducted in a real setting. This allowed us to obtain the first-hand experience of the facilities available with these firms. Additionally, we saw the products designed by them and had access to their documents, such as sketches, photos, and design drawings. This also allowed us to dynamically react to the responses and further probe, if required, through open-ended questions—something not possible in the design experiments and questionnaire analyses that are usually conducted in lab settings. Similarly, conducting diary studies or 43 observational studies in real settings for 24 firms would have been time intensive.

The interview method—which is widely used in design studies [35]—was used in this work to explore how designers elicit customer needs; how they formulate requirements for products; how they generate, evaluate, and select design concepts; and how they specify product details. Interviews took place between March 2021 and January 2022.

Random-sampling technique was adopted and used to select 24 informal metalworking microenterprises in the Coast and Dar es Salaam regions of the United Republic of Tanzania. The microenterprises in which interviews were conducted were randomly selected to avoid biases and to ensure that the eventual findings approximated those of the actual population. All the interviewed microenterprises were informal metalworking microenterprises, with no more than ten permanent employees. Direct communication and chain-referral sampling approaches [35] were used to recruit interview subjects. Table 1 presents the information on the selected microenterprises and the subjects who participated in the interviews. The major job orders that informal metalworking microenterprises receive include supplying building construction resources (e.g., aluminium windows, door frame grills, aluminium door frames, steel gates, doors, movable kiosks, office space partitioning, and window grills), metal furniture (e.g., bed frames, reading table frames, dressing tables, chair/couch frames, and TV stands.), household utilities (e.g., cooking utensils, charcoal cooking stoves, and charcoal grill stove), agricultural equipment (e.g., chicken feeding utensils), and light machinery (e.g., grain shelling machines, flour milling machines, and brick-making machines)—see Table 1. Figure 1 shows examples of products produced in studied microenterprises.

Table 1. Information on informal metalworking microenterprises and on interviewed participants.

		Firm Attributes			Information on Participants (Two from Each Firm)		
Firm		**Firm Age (Years)**	**Products Manufactured**	**Number of Staff**	**Education**	**Age (Years)**	**Total Experience (Years)**
A	1	2	Agricultural machines, metal windows, door frames, etc.	4	Bachelor's degree (engineering) and O level	27, 35	4, 1
B	2	3	Agricultural machines, door frames, etc.	3	Bachelor's degree (engineering) and vocational training	25, 20	3, 2
C	3	20	Furniture, tables, chairs, beds, and couches	4	Vocational training and STD II	30, 24	8, 3
D	4	9	Furniture (chairs and bed frames), doors, shoe stands, and window grills	5	Both subjects—STD VII	26, 24	8, 2
E	5	15	Door and window grills and furniture	3	Both subjects—STD VII	32, 22	15, 3
F	6	5	Aluminium windows, doors, and aluminium-frame furniture	6	Both subjects—STD VII	25, 28	8, 2
G	7	6	Charcoal stoves, barbeque ovens, water gutters, and metal suitcases	2	Both subjects—STD VII	30, 22	9, 4
H	8	6	Bed frames, door grills, and steel gates	2	Bachelor's degree (business administration) and STD VII	32, 20	9, 1
I	9	2	Door and window grills, bed frames, chair/couch frames, and movable kiosks	4	Both subjects—STD VII	22, 21	3, 2
J	10	2	Chicken feeding utensils, charcoal stoves, and metal suitcases	2	Both subjects—STD VII	30, 15	7, 1
K	11	3	Bed Frames, window grills, door grills, dressing table, gates, decoration stands, and dressing tables	7	STD VI and STD VII	28, 23	6, 2
L	12	11	Window grills, door grills, and fence gates	3	STD VI and STD VII	22, 28	3, 10
M	13	8	Gates, window grills, furniture—bed frames, table frames, chairs	4	STD VI and STD VII	23, 32	3, 8
N	14	8	Window grills and furniture—bed frames, table frames, chairs, etc.	4	VETA and STD VII	35, 30	10, 9
O	15	6	Aluminium windows and doors	8	BA degree—public relations and form IV	32, 27	5, 3
P	16	1	Widow and door grills, fence gates, and bed frames	4	Artisan (vocational training) and form IV	29, 23	15, 6
Q	17	7	Aluminium widows and door grills and partitioning of rooms	20	Form IV, diploma in public administration	20, 24	2, 5
R	18	6	Aluminium widows and door grills and aluminium wardrobes	4	Form IV and form IV	24, 27	5, 2
S	19	7	Flour milling machines, grain shelling machine, and grain processing machines	4	Bachelor degree in education and diploma in mechanical engineering	26, 51	5, 2
T	20	3	Door and window grills, bed frames, and fence gates	5	STD VII and STD VII	23, 22	7, 1
U	21	2	Aluminium profiled doors and windows, aluminium furniture, and deck rails	7	Form IV and form IV	33, 28	11, 7
V	22	9	Doors and windows grills, gates	4	Form IV and vocational training	18, 30	1, 4
W	23	20	Meat-grilling ovens, charcoal stoves, and gas stoves	4	STD VII and STD VII	36, 25	20, 6
X	24	19	Window and door frames, gate frames, cooking stoves, and railings	3	STD VII and STD VII	28, 25	10, 5



Figure 1. Examples of products produced in studied microenterprises.

Semistructured interviews [36,37] were conducted with two subjects who were familiar with design practices in the firm. One subject was the main speaker and the other was there to corroborate the accounts given by the main speaker and to provide clarifications or any additional information whenever required. As such, a total of 48 subjects participated in interviews. It should be noted that this was qualitative research and the intention was not to gather data or information for statistical analysis. The focus was, rather, on acquiring a proper understanding of the actual design practices through qualitative research. Only 5 of 48 interview participants held bachelor's degrees. All respondents were male, and most of them lacked technical or design training.

Interviews took place at the interviewees' places of work. This allowed the interviewers to informally observe the working practices and culture in informal metalworking microenterprises. The interviewees were asked to refer to particular projects during interviews. Figure 2 shows examples of working practices and conditions in these microenterprises. The mean duration of the actual interviews, excluding briefing and debriefing, was 51 min. We sought consent of the subjects to participate in the study, to audio- and video-record the interviews, to take pictures, and to use gathered data/information in analyses and publications.



Figure 2. Some examples of working practices and conditions in informal metalworking microenterprises.

Interviews in all 24 microenterprises were administered by using the Swahili language and were audio-recorded. The recorded audio contents were then transliterated in Swahili and eventually translated into the English language. A general inductive and iterative approach was then used to analyse the translated transcripts [36]. The analysis was content-driven.

Sections 4 and 5 present the results of the analysis. Excerpts from the transcripts of the interviews are included to illustrate the findings. Some of the excerpts of the interviews have been edited for ease of comprehension and any additional information is included in brackets.

4. Identification of Consumer Needs

There is little knowledge about how informal metalworking microenterprises in developing countries elicit customer needs [38]. In this section, we present findings on how the designers in microenterprises elicit customer needs and how they formulate and organize the requirements—which form the first part of the research questions of this study.

Interviews were conducted in the firms listed in Table 1. Generally, design processes in these informal metalworking microenterprises entailed interacting with clients with a view to gathering their wishes and preferences with respect to functionality, dimensions, material type, aesthetic features, and cost of the product.

4.1. Elicitation of Needs

We explored how the needs of the customers are identified. Elicitation of needs in these microenterprises essentially entails conversion of tacit and subjective customer verbatim constructs into needs statements. Needs and requirements for products in these firms largely originate from interactions and one-on-one conversations with customers, as evidenced by the following sample interview excerpts.

> The client paid visit once to the [our] firm to press order, but we also visited the client to verify the space dimensions [Firm I, movable kiosk] . . . I interacted with the client straight away by discussing the picture that he brought and we also paid the visit to the client to take measurements. [Firm M, fence gate]

Some of the requirements originate from existing products. Customers use sketches, engineering drawings, and pictures of existing products to describe how they wish their products to be. Pictures are widely used as stimuli for discussion—refer to the interview excerpts below.

> The client came with pictures of an existing bed frame and we discussed and changed its appearance and added seating feature [Firm K, bed frame] . . . After seeing the pictures and based on our personal experiences we recognized the needs [Firm L, gates and window grills] . . . the client used the picture to explain how she wanted the dressing table to be like [Firm N, dressing table] . . . the client brought a picture of window grill with decoration features he wanted". [Firm X, Window grill]

The designers and their clients refer to the pictures during discussions and agree on additional features to incorporate in the final designs.

For products with novel features, the design processes start from scratch. Some microenterprises use few requirements to produce trial products that they showcase to potential customers. Designers and customers use these trial products as stimuli for discussing how the product should be—see quotes below.

> For this product we developed a [sample] product and interacted with potential clients mostly during exhibitions [to explore its acceptability]. We visited several exhibitions to showcase the prototype. [Firm C, foldable multipurpose furniture]

Some designers visit clients and interview them at their sites—see sample excerpt below. The number of outgoing and incoming for such visits vary from two to ten. This allows the designers to know the use environment of the product.

> The client paid visit once to the (our) firm to press order, but we also visited (the client) to verify the space dimensions. [Firm I, movable kiosk]

The effectiveness of needs elicitation usually depends on the technique adopted and used [39–45]. Informal metalworking microenterprises use unstructured interview methods and appear not to prefer using formal alternative needs elicitation channels such questionnaire surveys, structured interviews, and maintenance reports.

4.2. Interviews with Customers

Interviews with customer is the approach that is widely used to collect needs data in informal metalworking microenterprises. Some researchers claim that interviews—particularly structured—are one of the most effective needs elicitation techniques [39]. Interviews enable the informal metalworking firms to get glimpses into how customers wish the product to function and be—see the representative excerpt below.

> We discussed and agreed with the client on how the gate should be like and how different [the client] wanted it to be compared to the pictured one. [Firm Q, aluminium windows]

Generally, it can be said that the needs originated from interviews with customers, which generated first-hand data that are used in the formulation of requirements. During conversations with clients, firms typically record first-hand needs data by taking notes, sketching concepts, or taking photos, as substantiated by the sample quotation below:

> We took notes [. . .] typically hand-written text complemented with hand sketches to describe concepts. [Firm A, palm oil filter]

However, as demonstrated by the excerpts below, some firms do not document conversations with clients and only just listen and recollect what was said later on.

> . . . . . we just heard and recalled what the clients had to say. [Firm J, chicken feeding utensil] . . . We didn't sketch anything, . . . we just got word-of-mouth explanations on the needs. [Firm G, Charcoal stoves, barbeque ovens, water gutters, metal suitcases]

Generally, the designers typically receive verbal explanations from clients but do not archive the conversations. They regularly use hand-written text, pictures, or hand-prepared sketches to describe requirements—see the representative excerpts below.

> [Customer came] with the sample photo or sketch. [Firm B, peanut peeling machine] . . . past experiences . . . helped us to identify requirements, and we then prepared sketches of the gate. [Firm T, fence gate]

Traditionally, in order to conduct meaningful needs analysis, it is imperative to properly document conversations. Documentation can be in the form of textual interview transcripts complemented with hand-sketches to record nonverbal information [42].

4.3. Sources of Requirements

In informal metalworking microenterprises, requirements originate from different sources. Apart from interviews, standards and use of past experiences are also the sources of requirements in many microenterprises. Some requirements also originate from design constraints and from the designer's own expert knowledge. Reusing past knowledge and experiences seems to be a natural way to elicit requirements and of handling uncertainties in informal metalworking microenterprises. Moreover, the requirements gathered in informal metalworking microenterprises describe technological, social, environmental, and economic aspects of the product.

4.4. Prioritization of Requirements

Designers in informal metalworking microenterprises identify, at most, eight requirements. The identified requirements describe functionality, appearance, dimensions, and

cost of the product, but are generally neither properly worded and documented, nor archived. The requirement statements do not properly describe the product.

Customers are involved in prioritizing the requirements. Apparently, the requirements that describe dimensions, appearance, cost, and functionality of products are given higher priority—see some of the representative excerpts below.

> The appearance and dimensional specifications were given higher priority [Firm N, dressing table]. The strength of materials and dimensional requirements were given higher priority. [Firm W, meat grilling oven]

However, some microenterprises do not prioritise requirements, but give equal priority to all identified requirements.

> All requirements were given equal priorities. [Firm P, windows grill]

In engineering design, requirements are typically prioritized systematically by using formal methods such as Quality Function Deployment (QFD) and Analytic Hierarchy Process (AHP) to help designers set definite goals for their products and to ensure that the design project focuses on and addresses the key customer's needs [39,42,43]. Our findings reveal that the informal metalworking microenterprises did not employ such methods.

4.5. Difficulties Faced in Identifying Needs

Some designers in informal metalworking microenterprises felt that the process of identifying needs is tedious and expensive. Some excerpts from the interviews to explain the difficulty faced are presented below.

> Our approach entailed making prototype and demonstrating concept. . . . was tedious . . . . [expensive], and . . . [lengthy]". [Firm C, foldable multipurpose furniture]

Other difficulties faced include the identification of needs that cannot be met due to limitations of capabilities of manufacturing equipment within the microenterprises, and customers disputing some of the obvious requirements, e.g., those based purely on technology-related and economic reasons—see some of the interview quotations below.

> The customer was [. . .] disputing technical requirements [. . .] wanted the product to be painted without spraying red-oxide paint first to cut cost . . . . [Firm D, door gate]

It was also observed that designers in informal metalworking microenterprises also confront numerous inherent difficulties typically faced in identifying customer needs and requirements. These include ambiguity and imprecision of requirement statements [40], uncertainty of whether needs are genuinely captured [44], and excessive focus on the technical details of a product [41].

5. Conceptual Design

In a formal design process, requirement identification is followed by a conceptual design stage. According to Pahl and Beitz [45]'s model of designing, the conceptual design stage involves creating function structures, identifying working principles, and combining the working principles into a working structure. Additionally, conceptual design focusses on novel idea generation.

Researchers converge on the view that *novelty* is a measure of the newness of a product with respect to existing products in the market satisfying the same function [46]. For instance, the first pin-hole camera, the first X-ray machine, and drugs such as penicillin are regarded as very highly novel products [47].

Industries are under huge pressure to launch novel products because of reasons such as increased competition and customer expectations, rapidly changing technology, and shorter product life cycles [48,49]. To generate novel designs, designers use methods such as brainstorming, biologically inspired design, TRIZ, functional analysis, etc. [50]. Apart from these methods, designers take inspiration from sources such as patents, expert opinions,

discussions with colleagues, and past experiences [51,52]. Greater product novelty, in turn, positively influences the product quality, which in turn determines the market share of a product [53–56]. Therefore, in formal design methods, novel idea generation is central to conceptual design.

This contrasts with the informal design process in which the decisions are driven largely by intuition and past experiences. In addition to following informal design processes, the microenterprises interviewed in this study functioned within constrained resources and limited technical proficiency. These factors are likely to influence the design outcomes. Overall, this section, describes the findings on how these firms generate and evaluate alternative product concepts, and how they define product details—which form the second part of the research questions of this study. More specifically, Sections 5.1–5.3 present the findings of our investigation with regard to the conceptual stage of the design process followed in the informal metalworking firms in Tanzania. Section 5.4 focuses on methods adopted by these firms for material selection and determination of dimensions.

5.1. General Approach for Concept Generation

We intended to abstract a general approach to concept generation, which the microenterprises in Tanzania follow. The employees of the firms were asked to brief the steps that they followed after requirement formulation. Sixteen firms responded that they manufactured the product straight away after requirement formulation. Six firms reported generating concept sketches for the purpose of communicating the design to the clients. We present some excerpts from the interview as evidence.

> I started to manufacture the sample product straight away after knowing the requirements and dimensional specifications. [Firm F, Aluminum windows, doors, aluminum-frame furniture]

> We proceeded to take measurements at the site, calculate costs, purchase raw materials and then we started to manufacture the product. [Firm R, Windows, door grills and wardrobes]

Clearly, this contrasts with the formal conceptual design process, which involves activities such as the exploration of concept space, evaluation of generated concepts, creation of prototypes, and a preliminary analysis of the proof of concept.

5.2. Concept Generation

We explored how designers in informal metalworking microenterprises generate concepts. Findings on how they explore and select concepts are presented below.

5.2.1. Exploration of Concepts

To understand whether the concept space was explored, we inquired about the number of alternative concepts generated. Twelve firms reported generating one concept and improvising it over multiple iterations after discussing with the clients. Three firms reported generating four concepts. Two firms reported generating six and nine concepts, respectively. Overall, the majority of the firms generated only one concept. Some excerpts from the interview to substantiate this are presented below.

> There were no any alternative concepts. The client wanted the kiosk to appear and be produced as the one shown in the picture that was provided. [Firm J, Chicken feeding utensils, charcoal stoves, metal suitcases]

> We prepared only one alternative concept and improved by continuously engaging the client" [Firm V, doors and windows grills, gates]

In design creativity research, the number of ideas generated (also known as idea fluency) directly correlates with the quality of ideas [57–60]. Design methods such as brainstorming support idea fluency and are widely used in industries [61]. As most of

these firms generated only one concept, it may be unlikely that these concepts would be of high quality.

We intended to understand if the firms had awareness about novelty and innovation. Most firms agreed that innovativeness involves new methods, new designs, the use of new materials, or the use of new technology. It can be inferred that these firms had an understanding that innovativeness involves newness. However, interview results revealed that they did not use scientifically established design methods for novel ideation. They depended on stimuli from social media sites, competitors' products, and discussions with colleagues. Only three firms reported using technical standards, engineering drawing and design methods such as brainstorming. When asked about the use of any special methods to design creative products, they did not report the use of any different methods.

5.2.2. Concept Selection

In a formal design process, designers use the established methods for concept selection. Concept evaluation methods, such as Pugh's method and rank-ordering method, use requirement satisfaction as criteria to evaluate concepts.

In the case of these firms, our observations were the following. Eleven firms reported that the concept was selected based on discussions with the client. Six firms said that concept selection was not required, as they generated only one concept and iterated over it. Two firms seemed to be aware of subfunction concepts and stated that they selected concepts based on discussions with the end users. Here are some related excerpts from the interview.

> For various sub-functions, there were alternative concepts, from which best were selected through discussions that involved both potential end users, and the final composition of the entire product concept was eventually generated. [Firm D, Furniture, doors, shoes-stands, window grills]

> By involving the client (who had the final say) and based on our past experiences. [Firm X, window and door frames, gate frames, cooking stoves, railings]

Furthermore, we intended to understand which product attributes they deemed as important for concept selection. Nine firms said that functionality was the key attribute for selection of concepts, seven firms reported that aesthetics was the key attribute for concept selection, while four firms reported durability as an important attribute.

5.3. Concept Representation

The concepts are represented using various modes, such as by using sketches, clay models, paper models, and CAD models. These representations serve various purposes, such as active learning, refinement, communication, and exploration [62]. From the interviews, it appeared that, in the cases of the interviewed firms, the concepts were represented primarily to communicate the design to the client and to take the approval for manufacturing. Concept representation seemed optional, and the ones that did mostly used hand-drawn sketches as the representation medium. Nine firms said that they manufactured the product directly without representing it, seven firms reported using concept sketches, and one firm reported making CAD models. Related excerpts are presented below.

> We arrived directly at the final solution—based on past experiences. [Firm T, Door and window grills, bed frames, fence gates]

> We directly manufactured the product according to the agreed dimensions and other requirements. [Firm U, Aluminium profiled doors and windows, aluminium furniture, deck rails]

Most firms did not formally document the sketches. Upon request, we collected sketches from three firms only (See Figure 3). It can be seen that the people who prepared most of these sketches lack the skills required for industrial sketching.

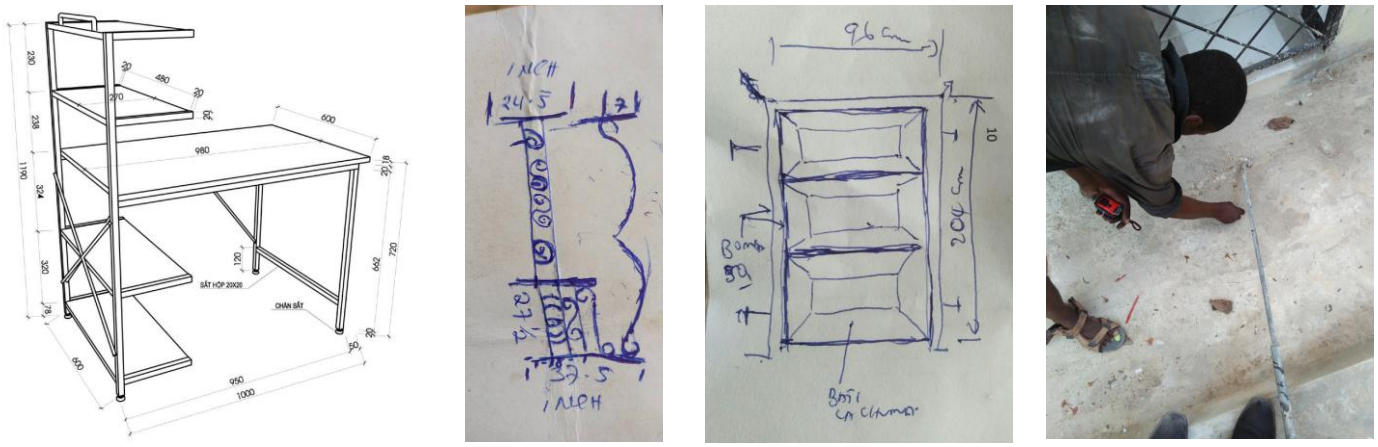


Figure 3. Examples of concept sketches produced by some of the firms interviewed in this study. (a) Firm D manufactures furniture such as chairs, bed frames, doors, shoe stands, and window grills. (b) Firm E manufactures door and window grills and furniture; employee sketching the concept on floor. (c) Firm I manufactures bed frames, door grills, and door gates.

5.4. Material Selection and Dimension Identification

With regard to main considerations for material selection, most firms used square hollow sections of steel and aluminium. Material cost, strength, durability, and availability were found to be the main considerations for material selection. The choice of material was made by the client. With regard to calculation of dimensions, some dimensions were established by visiting the site, while the other dimensions were determined based on past experiences. One firm also reported that they arrived at the dimensions by studying the competitor's product.

In a formal design process, material cost, strength, and availability are some of the governing criteria for material selection. Scientific methods and tools of material selection, such as the Cambridge Engineering Selector (CES) toolkit are widely used for material selection. None of the firms reported using such methods. Similarly, for the calculation of product dimensions, designers used methods such as quality function deployment [62], and referred to anthropometric data [63], industrial standards, and government regulations. However, in our study, only one firm reported using ergonomic standards.

6. Discussion

6.1. Needs Identification

Traditionally, needs elicitation in large formal enterprises is a well-organized process [64–66], which passes through definition of scope, raw data gathering, derivation of needs statements, and arrangement of needs according to their importance.

We explored how designers in informal metalworking microenterprises elicit consumer needs and requirements for products. We found that designers in these firms mainly use interview methods to identify needs and requirements for products. Overall, the designers only use their intuitions to identify the needs and are generally neither unaware of the existence of a formal process for eliciting needs nor of the tools and methods of identification of needs and requirements. Furthermore, we found that needs are not systematically interpreted and translated into requirements. Specifications are not quantified through a formal process that traditionally entails identifying metrics and measurement units, which corresponds with the needs or requirements. Specifications are not necessarily tied to the needs and requirements. The dimensional specifications for the products in these firms are established based on past experiences or by taking actual measurements from the sites. Other specifications are also determined in an ad hoc fashion.

Product specifications are, in fact, the product attributes or design features [67]. Formulation of specifications for a product is one of the mainstream early-stage design tasks,

through which concrete specifications are determined based on customer needs [68]. Quality function deployment (QFD) [69] is commonly used to determine the specifications for the products based on the needs. QFD utilizes house of quality (HoQ) to map customer needs and requirements to the specifications [70]. Other methods used include semantics methods [71,72], which apply.

Other identified shortcomings in needs elicitation include (1) the absence of mission statements; (2) assumptions that constrain the development efforts; (3) excessive reliance on the one-on-one interview method, which is known to have inherent drawbacks such as biases and reliance on interviewer capability [33,37]; (4) the absence of needs data that describe sensory experiences, such as comfort or style; and (5) ambiguity in the needs statements.

6.2. Conceptual Design

We explored how informal microenterprises generate and evaluate alternative design concepts and define product details. We found that the number of concepts generated by informal metalworking microenterprises were limited. In most cases, only one concept was generated, which was iterated over multiple cycles of improvement. We propose the following reasons for the generation of a limited number of concepts.

1. In eight cases, the clients brought the photo or the sketch of the final product. In some other cases, clients chose the product from the range of products that had been produced earlier by that firm. So, the firms already knew the product that they had to manufacture. The following excerpts convey this claim. There were no any alternative concepts. The client wanted the kiosk to appear and be produced as the one shown in the picture that was provided. [Firm J, Chicken feeding utensils, charcoal stoves, metal suitcases] Decoration features—the client wanted the bed frame to appear as in the picture. [Firm W, meat grilling ovens, charcoal stoves, gas stoves] The client visited us and selected the product that closely match the envisioned product. [Firm H, Bed frames, door grills, steel gate]
2. In our view, products such as window grills, gates, and metal frames received as orders by these firms are already highly explored by designers. The products that can challenge creative thinking and push the boundaries of technical know-how were usually not received as orders—as can be seen in Table 1. Therefore, these firms might not have felt the need to generate multiple concepts.
3. It appears from the interviews that the clients had the final say in the decision making. It is possible that clients, per se, were biased towards the existing designs and might not have been confident in the new concepts produced by these firms. Thus, this would have stopped these firms from generating new concepts.

Designers use various concept-representation modes, such as paper models, cardboard models, sketches—digital and physical, clay models, and CAD models [62]. For these modes of representation, concept sketches seemed to be the dominant method for concept representation in the case of the interviewed microenterprises. These sketches were informally documented with people who seemingly lacked the skills needed for industrial design sketching. Overall, the limited use of concept-representation methods might be due to lack of proper training and required resources. Further, it might be that the designers had prior experience with the manufactured products, so they did not feel the need to represent these designs in a detailed manner.

While scientifically established design methods such as SCAMPER and brainstorming were not used, the major sources of stimuli that they reported were social media sites, discussions with colleagues, and competitors' products. Again, this indicates a lack of training, awareness, and required skills for a designer.

Functionality and aesthetics seemed to be important product attributes for them. They determined critical dimensions by taking measurements from the use environments of the envisioned products, by using their own past experiences, or by adopting dimensions of the existing products. Overall, materials are selected by the designers, but customers

have the final say. The customer also has the final say in deciding on features of the product. Additionally, those surveyed appeared to be familiar with the concepts of novelty and innovation.

Our results corroborate earlier research, such as in [8,29], which reported that informal sectors avoid exploratory activities such as prototyping or tinkering due to costs and available resources. Additionally, researchers have emphasized on the need of co-designing at every phase of design process, especially for sustainable impact on marginalized societies [73]. In our study, we observed that these firms maintained regular interactions with their customers, who seemed to have the final say in the design and manufacturing of the product. This points towards the existence of co-designing practices in the informal sector.

7. Conclusions

This research investigated the design process of the informal microenterprises in Tanzania—in particular, the need identification, requirement formulation, and conceptual design phases of a design process. For this, we interviewed 24 metalworking enterprises at their workplaces in real settings. With regard to need identification and requirement formulation, it was found that these firms follow their intuitions to identify the needs and are generally neither unaware of the existence of a formal process for eliciting needs nor of the tools and methods of identification of needs and requirements. With regard to conceptual design, it was found that concept exploration and ideation methods are not followed. In most cases, they receive routine job orders, for which ideation in not considered to be as important. For both, need identification and conceptual design, these firms rely heavily on the past experiences and in general lack the necessary tools and training.

Additionally, this research has revealed that design activities, e.g., activities associated with needs identification and conceptual design, are influenced by the context in which those activities are performed. For instance, the informal metalworking microenterprises studied in this research face various constraints, such as a lack of formal design education and weak access to various resources. These sociocultural and economic aspects of the context have an influence on their design activities.

A limitation of this work is that the questionnaire used was written in English and translated into the Swahili language. The transcriptions of the responses in the Swahili language were translated into English. It is probable that these translations might have altered the meaning of some words.

This research raises several interesting questions for further research, some of which are discussed below.

1. It was observed that these firms, in most cases, used informal design methods. A better understanding is required on the effectiveness of formal design methods in the constrained environments, such as those in the interviewed firms. This is an interesting and important area for further research, as formal design methods are typically developed to support designers in contexts that are distinctly different from those in which the informal firms operate.
2. Further research can also aim to develop bespoke design methods for these types of firms, taking into account their educational background and the various resources that are available to them. These design methods can also be created with the firms, and then systematically tested to assess their impact on the design outcomes of the firms.
3. The employees in these firms maintained regular interactions with their customers, who seemed to have the final say in the design and manufacturing of the product. Involving customers in designing and manufacturing is a recent practice in industries, which comes under the purview of co-designing. It is unlikely that the employees of these firms, who did not have access to proper training, were aware of "co-design" as a recent practice. This points to something called *tacit co-design*, meaning that these firms unknowingly co-designed products in their own intuitive ways.
4. Most of these firms reported using the Internet through mobile phones only. Furthermore, only a few employees in these informal microenterprises had knowledge

of English. This calls for the necessary development of smartphone-based design methods and training programmes in local languages.

Overall, this research paves the way for a number of future research avenues in terms of developing dedicated design models and methods for constrained environments, studying tacit co-design and development of low-cost design tools in local languages. Also, it is likely that the results from this study hold true for resource-constrained enterprises in other developing countries.

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References

1. Papalambros, P.Y. Design Science: Why, What and How. *Des. Sci.* **2015**, *1*, 1–38. [CrossRef]
2. Jagtap, S. Design and poverty: A review of contexts, roles of poor people, and methods. *Res. Eng. Des.* **2019**, *30*, 41–62. [CrossRef]
3. Cross, N. *Designerly Ways of Knowing*; Springer: Berlin/Heidelberg, Germany, 2006.
4. Simon, H.A. *The Sciences of the Artificial*; MIT Press: Cambridge, MA, USA, 1996.
5. Ulrich, K.T.; Eppinger, S.D. *Product Design and Development*; McGraw-Hill: New York, NY, USA, 2007.
6. Subrahmanian, E.; Reich, Y.; Krishnan, S. *I Are Not Users: Dialogues, Diversity, and Design*; MIT Press: Cambridge, MS, USA, 2020.
7. Williams, C.C.; Shahid, M.S.; Martínez, A. Determinants of the level of informality of informal micro-enterprises: Some evidence from the city of Lahore, Pakistan. *World Dev.* **2015**, *84*, 312–325. [CrossRef]
8. Donaldson, K.M. Product design in less industrialized economies: Constraints and opportunities in Kenya. *Res. Eng. Des.* **2006**, *17*, 135–155. [CrossRef]
9. Kabecha, W. Technological capability of the micro-enterprises in Kenya's informal sector. *Technovation* **1998**, *19*, 117–126. [CrossRef]
10. Jagtap, S.; Larsson, A.; Hiort, V.; Olander, E.; Warell, A.; Khadilkar, P. How design process for the base of the pyramid differs from that for the top of the pyramid. *Des. Stud.* **2014**, *35*, 527–558. [CrossRef]
11. Nichter, S.; Goldmark, L. Small firm growth in developing countries. *World Dev.* **2009**, *37*, 1453–1464. [CrossRef]
12. Snyder, H. Literature review as a research methodology: An overview and guidelines. *J. Bus. Res.* **2019**, *104*, 333–339. [CrossRef]
13. Blessing, L.T.; Chakrabarti, A. *DRM: A Design Research Methodology*; Springer: London, UK, 2009.
14. Dorst, K.; Cross, N. Creativity in the design process: Co-evolution of problem-solution. *Des. Stud.* **2001**, *22*, 425–437. [CrossRef]
15. McMahon, C.A. Reflections on diversity in design research. *J. Eng. Des.* **2012**, *23*, 563–576. [CrossRef]
16. Cross, N.; Dorst, K.; Christiaans, H. *Analysing Design Activity*; Wiley: New York, NY, USA, 1996.
17. Chakrabarti, A.; Blessing, L.T. *Anthology of Theories and Models of Design*; Springer London Limited: London, UK, 2016.
18. Jagtap, S. Codesign in resource-limited societies: Theoretical perspectives, inputs, outputs and influencing factors. *Res. Eng. Des.* **2022**, *33*, 191–211. [CrossRef]
19. Webb, J.W.; Bruton, G.D.; Tihanyi, L.; Ireland, R.D. Research on entrepreneurship in the informal economy: Framing a research agenda. *J. Bus. Ventur.* **2013**, *28*, 598–614. [CrossRef]
20. Hart, K. Informal income opportunities and urban employment in Ghana. *J. Mod. Afr. Stud.* **1973**, *11*, 61–89. [CrossRef]
21. Kistruck, G.M.; Webb, J.W.; Sutter, C.J.; Bailey, A.V. The double-edged sword of legitimacy in base-of-the-pyramid markets. *J. Bus. Ventur.* **2015**, *30*, 436–451. [CrossRef]

22. Page, J.; Söderbom, M. Is small beautiful? Small enterprise, aid and employment in Africa. *Afr. Dev. Rev.* **2015**, *27* (Suppl. S1), 44–55. [CrossRef]
23. Schneider, F.; Enste, D.H. *The Shadow Economy: An International Survey*; Cambridge University Press: Cambridge, UK, 2013.
24. Thorp, R. Hernando De Soto, The Other Path: The Invisible Revolution in the Third World (London: IB Tauris, 1989). *J. Lat. Am. Stud.* **1990**, *22*, 403–405. [CrossRef]
25. ILO—International Labour Organization. *Women and Men in the Informal Economy: A Statistical Picture*, 2nd ed.; ILO: Geneva, Switzerland, 2013.
26. Brown, C.; Hamilton, J.; Medoff, J.L. *Employers Large and Small*; Harvard University Press: Cambridge, UK, 1990.
27. Bureau, S.; Fendt, J. Entrepreneurship in the informal economy: Why it matters. *Int. J. Entrep. Innov.* **2011**, *12*, 85–94. [CrossRef]
28. Dijk, M.P.V.; Sverrisson, Á. Enterprise clusters in developing countries: Mechanisms of transition and stagnation. *Entrep. Reg. Dev.* **2003**, *15*, 183–206. [CrossRef]
29. Bhalla, A. Innovations and small producers in developing countries. *Innov. Small Prod. Dev. Ctries.* **1989**, *24*, M2–M7.
30. Chuta, E.; Liedholm, C. Employment and Growth in Small-Scale Industry, Empirical Evidence and Policy Assessment from Sierra Leone. In *A Study Prepared for the ILO within the Framework of the World Employment Programme*; ILO: Geneva, Switzerland, 1985.
31. Sethuraman, S.V. *Technology Adaptation in Micro-Enterprises: The Case of Bangalore (India)*; ILO: Geneva, Switzerland, 1989.
32. Guimarges, L.E.; Penny, J.E.; Moody, S. Product design and social needs: The case of North–East Brazil. *Int. J. Technol. Manag.* **1996**, *12*, 849–864.
33. Müller, J. Befit for change: Social construction of endogenous technology in the South. In Proceedings of the FAU Conference-Workshop 4 on Community Entrepreneurs and Local Economic Development, Danhostel Copenhagen City, Denmark, 17–19 March 2010.
34. Cozzens, S.; Sutz, J. *Innovation in Informal Settings: A Research Agenda*; IDRC: Ottawa, ON, Canada, 2012.
35. Jagtap, S. *Co-Design with Marginalized People: Designers' Perceptions of Barriers and Enablers*; CoDesign: Miami, FL, USA, 2021; pp. 1–24.
36. Gray, D.E. *Doing Research in the Real World*; Sage: London, UK, 2013.
37. Breakwell, G.M. "Interviewing Methods". In *Research Methods in Psychology*, 3rd ed.; Breakwell, G.M., Hammond, S., Fife-Schaw, C., Smith, J.A., Eds.; SAGE: London, UK, 2006; pp. 232–253.
38. Opiyo, E.; Jagtap, S.; Keshwani, S. Conceptual Design in Metalworking Microenterprises: An Empirical Study in Tanzania. In Proceedings of the International Design Conference—Design 2022, Online, 27 May 2022; pp. 2493–2502. [CrossRef]
39. Davis, A.; Dieste, T.Ó.; Hickey, A.; Juristo, N.; Moreno, A. Effectiveness of Requirements Elicitation Techniques: Empirical Results Derived from a Systematic Review. In Proceedings of the 14th IEEE International Conference Requirements Engineering, St. Paul, MN, USA, 11–15 September 2006; pp. 176–185. [CrossRef]
40. Jiao, R.; Chen, C.-H. Customer Requirement Management in Product Development: A Review of Research Issues. *Concurr. Eng.* **2006**, *14*, 173–185. [CrossRef]
41. Chen, C.-H.; Khoo, L.P.; Yan, W. Web-enabled Customer-oriented Product Concept Formationvia Laddering Technique and Kohonen Association. *Concurr. Eng. Res. Appl.* **2002**, *10*, 299–310. [CrossRef]
42. Bolderston, A. Conducting a Research Interview. *J. Med. Imaging Radiat. Sci.* **2012**, *43*, 66–76. [CrossRef] [PubMed]
43. Fukuda, S.; Matsuura, Y. Prioritizing Customers' Requirements for Concurrent Design. *Trans. Jpn. Soc. Mech. Eng. Ser.* **1994**, *60*, 3638–3642. [CrossRef]
44. Yan, W.; Chen, C.-H.; Khoo, L.P. An Integrated Approach to the Elicitation ofCustomer Requirements for Engineering Design using Picture Sorts and Fuzzy Evaluation. *AIEDAM* **2002**, *16*, 59–71. [CrossRef]
45. Beitz, W.; Pahl, G.; Grote, K. Engineering Design: A Systematic Approach. *Mrs Bull.* **1996**, *21*, 71.
46. Sternberg, R.J.; Lubert, T.I. The concept of creativity: Prospects and paradigms. In *Handbook of Creativity*; Sternberg, R.J., Ed.; Cambridge University Press: Cambridge, UK, 1999; pp. 3–15.
47. Sarkar, P.; Chakrabarti, A. Assessing Design Creativity. *Des. Stud.* **2011**, *32*, 348–383. [CrossRef]
48. Helfat, C.E.; Peteraf, M.A. The dynamic resource-based view: Capability lifecycles. *Strateg. Manag. J.* **2003**, *24*, 997–1010. [CrossRef]
49. Cinquini, L.; Di Minin, A.; Varaldo, R. (Eds.) *New Business Models and Value Creation: A Service Science Perspective*; Springer: Milan, Italy, 2013.
50. Chulvi, V.; Mulet, E.; Chakrabarti, A.; López-Mesa, B.; González-Cruz, C. Comparison of the Degree of Creativity in the Design Outcomes Using Different Design Methods. *J. Eng. Des.* **2012**, *23*, 241–269. [CrossRef]
51. Lee, S.; Lee, S.; Seol, H.; Park, Y. Using patent information for designing new product and technology: Keyword based technology roadmapping. *Rd Manag.* **2008**, *38*, 169–188. [CrossRef]
52. Lanzotti, A.; Carbone, F.; Grazioso, S.; Renno, F.; Staiano, M. A new interactive design approach for concept selection based on expert opinion. *Int. J. Interact. Des. Manuf. (IJIDeM)* **2018**, *12*, 1189–1199. [CrossRef]
53. McMullen, J.S.; Shepherd, D.A. Entrepreneurial Action and The Role of Uncertainty in The Theory of the Entrepreneur. *Acad. Manag. Rev.* **2006**, *31*, 132–152. [CrossRef]
54. Ames, M.; Runco, M.A. Predicting Entrepreneurship from Ideation and Divergent Thinking. *Creat. Innov. Manag.* **2005**, *14*, 311–315. [CrossRef]

55. Ottosson, S. Boosting Creativity in Technical Development. In Proceedings of the Workshop in Engineering Design and Creativity, Pilsen, Czech Republic, 16–18 November 1995; pp. 35–39.
56. Molina, A.; Al-Ashaab, A.H.; Ellis, T.I.; Young, R.I.; Bell, R. A review of computer-aided simultaneous engineering systems. *Res. Eng. Des.* **1995**, *7*, 38–63. [CrossRef]
57. Diehl, M.; Stroebe, W. Productivity Loss in Brainstorming Groups: Toward the Solution of a Riddle. *J. Pers. Soc. Psychol.* **1987**, *53*, 497–509. [CrossRef]
58. Paulus, P.B.; Kohn, N.W.; Arditti, L.E. Effects of Quantity and Quality Instructions on Brainstorming. *J. Creat. Behav.* **2011**, *45*, 38–46. [CrossRef]
59. Kudrowitz, B.M.; Wallace, D. Assessing the Quality of Ideas from Prolific, Early-Stage Product Ideation. *J. Eng. Des.* **2013**, *24*, 120–139. [CrossRef]
60. Kudrowitz, B.; Dippo, C. Getting to the Novel Ideas: Exploring the Alternative Uses Test of Divergent Thinking. In *Proceedings of the 25th International Conference on Design Theory and Methodology, Portland, OR, USA, 4–7 August 2013*; American Society of Mechanical Engineers: New York, NY, USA, 2015; Volume 7, p. V005T06A013.
61. Isaksen, S.G. *A Review of Brainstorming Research: Six Critical ISSUES for Inquiry*; Creative Research Unit, Creative Problem Solving Group-Buffalo: Buffalo, NY, USA, 1998.
62. Camburn, B.; Viswanathan, V.; Linsey, J.; Anderson, D.; Jensen, D.; Crawford, R.; Wood, K. Design prototyping methods: State of the art in strategies, techniques, and guidelines. *Des. Sci.* **2017**, *3*, e13. [CrossRef]
63. Roozenburg, N.F.; Eekels, J. *Product Design: Fundamentals and Methods*; Wiley: Hoboken, NJ, USA, 1995.
64. Davey, B.; Cope, C. Requirements Elicitation—What's Missing. *Issues Inf. Sci.* **2008**, *5*, 543–551. [CrossRef]
65. Opiyo, E.Z. An Approach to Represent and Communicate Product or System Design Ideas at the Fuzzy-Front End of the Design Process. *Systems* **2016**, *4*, 8. [CrossRef]
66. Opiyo, E.Z. Reconnoitring how structural design in the context of industrial design engineering evolves. *Int. J. Prod. Dev.* **2016**, *21*, 1–19. [CrossRef]
67. Fung, R.Y.K.; Popplewell, K.; Xie, J. An Intelligent Hybrid System for Customer Requirements Analysis and Product Attribute Targets Determination. *Int. J. Prod. Res.* **1998**, *36*, 13–34. [CrossRef]
68. Stauffer, L.; Morris, L. A New Program to Enhance the Development of Product Requirements. In Proceedings of the NSF Design and Manufacturing Systems Conference, Atlanta, GA, USA, 8–10 January 1992.
69. Prasad, B. Review of QFD and Related Deployment Techniques. *J. Manuf. Syst.* **1998**, *17*, 221–234. [CrossRef]
70. Clausing, D. *Total Quality Development: A Step-by-Step Guide to World Class Concurrent Engineering*; ASME Press: New York, NY, USA, 1994.
71. Shoji, S.; Graham, A.; Walden, D. *A New American TQM*; Productivity Press: Portland, OR, USA, 1993.
72. Morris, L.J.; Stauffer, L.A. A Design Taxonomy for Eliciting Customer Requirements. In Proceedings of the 16th Annual Conference on Computers and Industrial Engineering, Pergamon, New York, NY, USA, 1 September 1994.
73. Rivett, U.; Marsden, G.; Blake, E. ICT for development: Extending computing design concepts. In *Africa-Centred Knowldges: Crossing Fields and Worlds*; Cooper, B., Morrell, R., Eds.; Boydell & Brewer Ltd.: Suffolk, UK, 2014; pp. 126–141.

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